SHOWCASE PORTFOLIO · NEURO-ENGINEERING

MARVIN NGABONZIRA

Neuro-Engineering Researcher & Science Communicator · Founder, The Science Blueprint

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01 PROFILE

Biomedical & Mechatronics Engineering student at Kyambogo University obsessed with **the gray area between biological brains and silicon machines**. Transitioned from maintaining clinical diagnostic hardware to reverse-engineering intelligence itself. Founder of ***The Science Blueprint*** — a science communication brand dedicated to decoding Brain-Computer Interfaces, neuro-engineering, and the mechanics of human cognition. Combines rigorous independent research with editorial-quality science storytelling. Currently building toward graduate research in South Korea.

02 RESEARCH & PUBLICATIONS

TSB-TR-2026-001 · TECHNICAL REPORT

Neural Data Privacy and Emerging Surveillance Risk of Brain-Computer Interfaces

Zenodo · DOI: 10.5281/zenodo.19058469 · March 2026 · Available: ResearchGate

Examines the emergent threat landscape at the intersection of consumer-grade BCI adoption and neural data harvesting. Analyzes surveillance vectors through a fictional company case study (MindSync Corp.), proposes a layered framework for neural privacy by design, and assesses regulatory gaps in existing data protection legislation.

TEMB 3201 · GROUP UNDERGRADUATE PROJECT

Adaptive Motor-Imagery BCI Using Riemannian Geometry (A-MDRM System)

Kyambogo University · Dept. of Mechatronics & Biomedical Engineering · 2026 (In Progress)

Developing a session-adaptive EEG Motor Imagery classifier leveraging Riemannian geometry on covariance matrices for subject-independent decoding. Implements MOABB BCI Competition IV Dataset 2a pipeline with Arduino hardware-in-the-loop validation. Target venues: Journal of Neural Engineering or IEEE TNSRE.

INDEPENDENT RESEARCH FOCUS

Brain-Computer Interfaces · Neural Data Privacy · Riemannian Geometry in ML · Neuromorphic Computing · Von Neumann vs. Biological Neural Architectures · Organoid Intelligence

03 SCIENCE COMMUNICATION

Founder & Lead Creator

THE SCIENCE BLUEPRINT · "REVERSE ENGINEERING INTELLIGENCE" · YOUTUBE & LINKEDIN · 2025 – PRESENT

- **Translating Complexity:** Scripts, produces, and edits long-form video essays dissecting cutting-edge BCI hardware (Utah Arrays, Neuralink N1), neural decoding algorithms, and the philosophy of mind — written for the "smart curious" audience with no neuroscience prerequisite required.
- **Dual-Format Research Pipeline:** Each content piece begins as a LinkedIn post (engagement layer) and expands into a formal technical report for Zenodo and ResearchGate — building a visible, citable research output record as an undergraduate.
- **Technical Storytelling:** Bridges engineering precision and narrative drive via split-screen visual aids, hypothesis-first hooks, and story structures modeled on communicators like Cleo Abram, Veritasium, and Neil deGrasse Tyson — adapted for the neuro-engineering niche.

04 EDUCATION

BSc. Biomedical & Mechatronics Engineering

KYAMBOGO UNIVERSITY · KAMPALA, UGANDA · 3RD YEAR · EXPECTED 2027

Dept. of Mechatronics & Biomedical Engineering. Core focus on the integration of biological systems with electromechanical design, signal processing, and control theory.

KEY MODULES:

Bio-Instrumentation · Control Systems · Neural Engineering · Mechatronics Design · Signal Processing · Embedded Systems · Biomedical Sensors

Diploma in Biomedical Engineering

KYAMBOGO UNIVERSITY · KAMPALA, UGANDA

Foundational programme with hands-on clinical hardware focus. Gained practical experience diagnosing, repairing, and maintaining complex hospital diagnostic systems — forming the operational baseline for the pivot from *maintaining* machines to **merging them with the human nervous system**.

05 TECHNICAL ARSENAL

HARDWARE & SYSTEMS

Arduino · Embedded C · Biomedical Instrumentation · Clinical Diagnostic Hardware · Mechatronic Systems Design · Sensor Integration

SOFTWARE & DATA

Python · MATLAB MNE-Python · MOABB Scikit-learn · NumPy · EEG Signal Processing · Riemannian Geometry (pyRiemann)

RESEARCH & WRITING

Technical Report Authorship · Publishing Literature Synthesis · Academic Citation · Formatting LaTeX (basic)

MEDIA & PRODUCTION

Scriptwriting & Narrative Structure · Science Communication · LinkedIn Content Strategy ·Video Editing

06 LANGUAGES

English	PROFESSIONAL / ACADEMIC
Luganda	NATIVE
Korean	ACTIVE STUDY — TOPIK II TARGET 2028

Korean actively self-studied in preparation for graduate study in South Korea (GKS Scholarship target, 2028).